

Figure F2: L1 Wide-Format Column Structure to Normalised Long-Form Schema Ingestion

| Wide-Format Table: Raw Chandrayaan-3 L1 CSV (2,055 Columns) |       |       |      |                                                        |           |     |           |
|-------------------------------------------------------------|-------|-------|------|--------------------------------------------------------|-----------|-----|-----------|
| Housekeeping & Metadata (6 Columns)                         |       |       |      | Spectral Wavelength Intensity Channels (2,049 Columns) |           |     |           |
| Time                                                        | Count | Reset | Fire | 164.35 nm                                              | 164.74 nm | ... | 878.26 nm |
| 11:35:22                                                    | 1     | 1     | 0    | 366.0                                                  | 384.0     | ... | 305.0     |

Melt / Wide-to-Long Reshape Ingestion Invariant



| Normalized Long-Format Table: spectral_data Schema (Rows × 2,094 per shot) |                    |                   |                |          |  |
|----------------------------------------------------------------------------|--------------------|-------------------|----------------|----------|--|
| Measurement_ID (PK)                                                        | Wavelength_nm (PK) | Cleaned_Intensity | Laser_Energy_V | Time_UTC |  |
| 1                                                                          | 164.35             | 0.0               | 3327           | 11:35:22 |  |
| 1                                                                          | 164.74             | 0.0               | 3327           | 11:35:22 |  |
| ...                                                                        | ...                | ...               | ...            | ...      |  |
| 1                                                                          | 878.26             | 3.0               | 3327           | 11:35:22 |  |